

ISC Class XII Maths — Chapter 6

Application of Derivatives — Formula Sheet

A. Rate of Change — Core Principle

If $y = f(x)$ and both x, y vary with time t , then by the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

More generally, if a relation connects several time-varying quantities (e.g. V, r, h), differentiate the relation *implicitly with respect to t* to connect their rates.

A rate is **positive** if the quantity is increasing, **negative** if decreasing — always state the sign in the final answer.

B1. Mensuration — Circle, Square, Cube

Circle (radius r): Area $A = \pi r^2$ Circumference $C = 2\pi r$

Square (side a): Area $A = a^2$ Perimeter $P = 4a$

Rectangle (sides l, b): Area $A = lb$ Perimeter $P = 2(l + b)$

Cube (side a): Volume $V = a^3$ Surface area $S = 6a^2$

B2. Mensuration — Sphere, Cylinder, Cone

Sphere (radius r): Volume $V = \frac{4}{3}\pi r^3$ Surface area $S = 4\pi r^2$

Cylinder (radius r , height h): Volume $V = \pi r^2 h$
Curved S.A. = $2\pi r h$ Total S.A. = $2\pi r h + 2\pi r^2$

Cone (radius r , height h , slant l): Volume $V = \frac{1}{3}\pi r^2 h$ $l^2 = r^2 + h^2$ Curved S.A. = $\pi r l$

Cone with semi-vertical angle α : $\tan \alpha = \frac{r}{h}$,
so $r = h \tan \alpha$ — substitute this to write V in terms of h alone.

Worked Pattern: Related Rates Problems

1. Identify all time-varying quantities and name them ($r, h, V, \dots$); note the given rate(s) and the rate asked for.
2. Write the geometric relation connecting the quantities (Sec. B1/B2), reducing to **one** independent variable using any given constraint (e.g. constant ratio, semi-vertical angle).
3. Differentiate the relation *implicitly with respect to t* .
4. Substitute the known values (including the value of the variable itself, found if necessary) and solve for the required rate.

Worked Pattern: Similar Triangles Setup

For a light source and a walking person, or height vs. horizontal distance problems (shadow-length, kite-string type):

1. Draw the figure; identify two similar right triangles (or one right triangle via Pythagoras for kite-string problems).
2. Write the proportion between corresponding sides (similar triangles), or $x^2 + h^2 = s^2$ (Pythagoras, kite at constant height h).
3. Differentiate with respect to t ; substitute known rates and values to solve.

C. Tangents & Normals — Core Formulas

Slope of tangent at (x_1, y_1) : $m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$

Slope of normal: $m_N = -\frac{1}{m_T}$ (normal $\perp$ tangent)

Equation of tangent: $y - y_1 = m_T(x - x_1)$

Equation of normal: $y - y_1 = m_N(x - x_1)$

Angle of inclination θ with the x -axis: $m = \tan \theta$

D. Tangents & Normals — Special Conditions

Tangent parallel to x -axis: $\frac{dy}{dx} = 0$ **Tangent parallel to y -axis** (normal horizontal): $\frac{dx}{dy} = 0$

Parallel to a line of slope m : $m_T = m$ **Perpendicular to a line** of slope m : $m_T \cdot m = -1$

Angle between two curves at a common point:
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Orthogonal intersection: $m_1 \cdot m_2 = -1$

Two curves touch at a point $\iff$ they have the *same* y -value **and** the *same* slope there.

Worked Pattern: Wavy Curve Method

To solve $f'(x) \gtrless 0$ where $f'(x)$ is a product/quotient of linear (or repeated) factors:

1. Factorise $f'(x)$ completely; mark all zeros of numerator *and* denominator on a number line (denominator zeros are always excluded from the domain).
2. Starting from the right of the largest root, mark +; cross each root right-to-left, **flipping** sign at roots of odd multiplicity, and **not flipping** at roots of even multiplicity (repeated/squared factors).
3. Read off the intervals where $f'(x) > 0$ (increasing) and $f'(x) < 0$ (decreasing) directly from the sign pattern.

Worked Pattern: Tangent & Normal Equations

1. If only x_1 (or only y_1) is given, substitute into the curve's equation to find the other coordinate.
2. Differentiate the curve to get $\frac{dy}{dx}$; evaluate at (x_1, y_1) to get m_T (implicit differentiation if the curve mixes x, y ; parametric formula $\frac{dy/dt}{dx/dt}$ if parametric).
3. Find $m_N = -1/m_T$.
4. Write both equations using $y - y_1 = m(x - x_1)$ with the respective slope.

E. Increasing & Decreasing Functions

f is **increasing** on interval I if $f'(x) \geq 0$ for all $x \in I$ (**strictly increasing** if $f'(x) > 0$).

f is **decreasing** on I if $f'(x) \leq 0$ for all $x \in I$ (**strictly decreasing** if $f'(x) < 0$).

Method: find $f'(x)$, factorise fully, then solve $f'(x) \geq 0$ and $f'(x) \leq 0$ using the sign scheme.

F. Maxima & Minima — Critical Points & Tests

Critical (stationary) point: $f'(c) = 0$ or $f'(c)$ does not exist.

First Derivative Test: if f' changes sign $+\rightarrow -$ at c , local **max**; if $-\rightarrow +$, local **min**; if no sign change, neither.

Second Derivative Test: at a point where $f'(c) = 0$: if $f''(c) < 0 \Rightarrow$ local **max**; if $f''(c) > 0 \Rightarrow$ local **min**; if $f''(c) = 0 \Rightarrow$ test fails, fall back to the first derivative test.

G. Absolute Maxima & Minima on $[a, b]$

1. Find all critical points of f inside the open interval (a, b) (solve $f'(x) = 0$).
2. Evaluate f at *every* critical point found **and** at both endpoints $f(a), f(b)$.
3. The **largest** value found is the absolute maximum; the **smallest** is the absolute minimum.

Tip: No second derivative test is needed for absolute extrema on a closed interval — just compare the raw function values from step 2 directly.

Worked Pattern: General Optimization Strategy

1. Read carefully and assign variable names to every quantity; note what is to be maximized/minimized and what is held **constant** (the constraint).
2. Use the constraint to eliminate one variable, writing the quantity to optimize as a function of a **single** variable (use Sec. B1/B2 for standard shapes).
3. Differentiate, set $= 0$, and solve for the critical value(s) within the valid physical domain (reject negative lengths, etc.).
4. Apply the second derivative test (Sec. F) to confirm max or min — or compare boundary values if the domain is a closed interval.
5. Substitute back to state the actual maximum/minimum value **and**, if asked, the dimensions that achieve it.

H. Common Traps to Avoid

- Every related-rates problem needs **implicit differentiation w.r.t. t** — never differentiate w.r.t. the spatial variable by mistake.
- Reduce to *one* variable using the given constraint *before* differentiating in optimization — differentiating a two-variable expression directly is a common error.
- Always check the **domain**: reject negative lengths/radii, and confirm a critical point actually lies inside the valid physical range.
- For "two curves touch," you need *both* equal y -values *and* equal slopes at the point — equal y alone only gives intersection, not tangency.
- In the wavy-curve method, a **repeated (even-power) factor** does not change sign at its root — forgetting this flips the entire sign pattern from that point onward.
- $f''(c) = 0$ does **not** mean "no extremum" — it means the second derivative test is inconclusive; go back to the first derivative test.

I. Quick Recognition Guide

Pattern seen	Use
“Rate of change of...”, two quantities linked by a formula, one rate given	Sec. A + Worked Pattern: Related Rates
Standard shape (circle, cube, sphere, cylinder, cone) dimensions changing with time	Sec. B1/B2 for the relevant formula, then differentiate w.r.t. t
Ladder, shadow, kite, or walking-person problems	Worked Pattern: Similar Triangles
“Find the slope of the tangent/normal at...”	Sec. C
“Find the equation of the tangent/normal...”	Worked Pattern: Tangent & Normal Equations
Tangent parallel/perpendicular to a given line or axis	Sec. D
Two curves given, “find angle between them” or “show orthogonal”	Sec. D angle/orthogonal formula
“Show the curves touch”	Sec. D — match y and slope
“Show f is increasing/decreasing” or “find intervals of monotonicity”	Sec. E + Worked Pattern: Wavy Curve
“Find local maxima/minima”	Sec. F, first or second derivative test
“Find absolute max/min on $[a, b]$ ”	Sec. G — compare all critical points and endpoints
Word problem: maximize/minimize area, volume, cost, perimeter subject to a constraint	Worked Pattern: General Optimization Strategy